

From Numerical Reproduction to Causal-Closure Boundaries

A Layered Audit of Public Hubble-Constant Inference Products

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Scope: synthesis of four frozen audit packages. No new H_0 estimate, causal explanation, DOI, repository record, or endorsement is asserted.

Abstract

Public cosmology releases increasingly provide tables, posterior samples, covariance matrices, notebooks, configuration files, and source code. The availability of these objects is often described collectively as “reproducibility,” although distinct scientific questions are answered by reproducing a headline number, preserving an inference under a change of representation, retaining information needed for model diagnostics, tracing an executed pipeline, or validating a generative model. This study consolidates two completed seasons of a public-product audit of Hubble-constant inference and a final internal validation program. The record covers a broad first-season dependency map and focused second-season case studies of gravitational-wave standard-siren posteriors, Pantheon+ bias-correction source readiness, the singular covariance used by the Local Distance Network (HODN), HODN supernova compression and same-object residual structure, and a separate TDCOSMO posterior-output cross-check. No new Hubble-constant estimate, physical model, or tension significance is introduced.

The synthesis uses seven working coordinates: artifact identity, numerical/output traceability, algebraic representation relation, target-inference invariance or sufficiency, diagnostic/model-support sufficiency, executed lineage/provenance, and generative/causal closure. These coordinates are a working audit framework induced by the audited cases, not a universal theory of reproducibility. Three results illustrate why the coordinates must be separated. First, frozen GWTC-4 and GWTC-5 v1 posterior files reproduce their published one-decimal summaries, while the published v1 uncertainty-reduction percentage is not recovered from the audited headline posterior pair; its exact comparator provenance remains open. Second, the HODN baseline is reproduced as $H_0=73.4987536436\pm0.8088000253 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Its design is nearly contained in covariance support, whereas the rounded public data vector has an off-support component with norm 0.188749 mag. When that off-support vector is evaluated using the project’s projected Moore–Penrose loss, an exact non-orthogonal row rescaling changes the target by $-0.0524454226 \text{ km s}^{-1} \text{ Mpc}^{-1}$; this does not imply non-invariance of an on-support singular Gaussian likelihood. Third, a 277-row supernova block is sufficient for the frozen fixed-covariance, one-intercept linear Gaussian target after scalar compression, yet the compression omits a parameter-independent residual term of $\chi^2=206.7606364$ for 276 degrees of freedom and therefore is not sufficient for residual diagnostics. The unexpectedly low same-name contrast statistic remains localized and

internally robust, but public evidence does not close the executed path from photometric inputs to final corrected rows.

The combined record contains no genuine cross-season scientific contradiction after aligning proposition, source product, numerical contract, and time. It supports bounded output-level and mathematical conclusions, identifies completed source-readiness HOLDS, and closes repeated internal analysis of the frozen evidence. It does not establish a corrected or preferred H_0 , a revised tension significance, a unique systematic cause, erroneous BBC corrections, a defective named pipeline, a solution to the Hubble tension, or new physics.

Keywords: Hubble constant; reproducibility; provenance; singular covariance; Moore–Penrose inverse; Type Ia supernovae; standard sirens; model diagnostics

1. Introduction

The empirical status of the Hubble constant is discussed through many observational routes, likelihood constructions, and released data products. Local-distance networks and calibrated Type Ia supernovae [1–4,11,12], baryon acoustic oscillations and cosmic microwave background analyses [5–8], megamaser and peculiar-flow constraints [9,10], gravitational-wave standard sirens [13–17], and strong-lensing time delays [18] answer related but non-identical questions and often share calibrations, assumptions, or data. A numerical disagreement between routes is scientifically important, but auditing the evidence behind those routes requires more than asking whether a published central value can be recomputed.

Several forms of agreement can coexist with important unresolved questions. A posterior file may reproduce a printed quantile while its generating likelihood is unavailable. An estimator may be stable under solver cutoffs in one coordinate representation yet change under an exactly equivalent non-orthogonal transformation. A compressed datum may preserve the likelihood for one parameter while discarding the residual structure required to assess model adequacy. Public configuration files may identify plausible inputs without proving which bytes were used in an executed production run. Treating all of these states as either “reproduced” or “not reproduced” obscures rather than clarifies the scientific record.

This paper reports a manuscript-level synthesis of a closed audit project. The project proceeded in two seasons. The first season built a broad map of dependencies and public-product traceability across local and early-Universe routes, DESI and CMB response directions, ACT public variants, TDCOSMO chains, gravitational-wave products, local-flow interfaces, and the HODN network. The second season revisited a narrower group of questions with fixed source products: GWTC-4/5 standard-siren posterior summaries, Pantheon+ BBC truth-closure readiness, singular-covariance inference in HODN, and the supernova block entering the HODN network. A cross-season audit aligned the claims, and a final internal validation package retested the highest-information numerical and logical mechanisms. The final state is PASS for the admitted internal validation contract and STOP for further reanalysis of the same frozen evidence.

The purpose is methodological and evidential, not cosmological parameter estimation. The study asks: what, precisely, has been reproduced; what information is preserved for the target inference; what information is lost for diagnostics; how far can public lineage be traced; and where must causal claims stop? The paper’s principal contribution is a seven-coordinate working framework and its application to a set of concrete public cosmology products. The framework is an audit map, not a claim that reproducibility universally has exactly seven dimensions.

2. Materials, source freeze, and audit contract

2.1 Frozen internal source record

The manuscript is derived from four immutable master packages:

1. *Hubble Tension First Season Master Reference Package*, version 1.0.0, SHA-256 3e6df9f . . . 952663c.
2. *Hubble Tension Second Season Master Reference Package*, version 1.0.0, SHA-256 cc15c96a . . . e964940.
3. *Hubble Tension Cross-Season Audit Package*, version 0.1.0, SHA-256 9719fae4 . . . 66e60cf.
4. *Hubble Tension Final Internal Validation and Closure Audit Package*, version 0.1.0, SHA-256 db70c27d . . . b95fb2f.

Each archive passed its bundled integrity verifier. The final validation package reported 24 of 24 numerical gates passed, a valid STOP marker, and consistency with the recorded source identities. Nested branch masters were checked against their registered hashes before manuscript drafting. Full hashes, archive member paths, result keys, and manuscript locators are supplied in the accompanying evidence register. Truncated hashes above are descriptive only; the register contains the authoritative full values.

2.2 What counts as evidence in this paper

The evidential unit is a proposition tied to a source product, numerical contract, and date or version. The audit did not substitute a nearby posterior, reconstruct missing dependence models, choose a rounding rule to force agreement, or infer an executed pipeline from filenames alone. When required inputs were non-unique or missing, the scientific result is a bounded source-readiness HOLD rather than an invented completion.

Project-internal reimplementations are described as alternate implementations, high-precision checks, or independent parsers within the project. They are not labeled external replications. “Independent” in this manuscript is reserved for independence of code path or parser inside the project and is always qualified accordingly. No outside research group was commissioned to reproduce the analyses.

2.3 Cross-season contradiction test

An apparent disagreement was counted as a scientific contradiction only if the earlier and later records addressed the same proposition, source/product, numerical contract, quantity, and time. Under this test, the cross-season ledger found zero true scientific contradictions. Later work sharpened some first-season statements—for example, fixed-representation stability of the HODN pseudoinverse solution was qualified by a representation test—but did not reverse the earlier bounded claim, which had applied only to the frozen representation and cutoff range.

2.4 Statistical conventions

Unless otherwise stated, numbers are quoted at the precision required to identify the frozen result. GW posterior summaries use the exact quantile convention registered by the branch audit: percentiles 15.865, 50, and 84.135 with Hyndman–Fan type-7 interpolation [26], followed by the specified one-decimal display rule. Generalized least-squares calculations use the covariance and solver policies stated in the relevant frozen package. The Moore–Penrose inverse is the unique generalized inverse satisfying the Penrose conditions [19]; singular-value methods provide a standard computational route [21]. Technical Supplement A gives the HODN formulation, and Technical Supplement B gives the fixed one-intercept, fixed-covariance supernova compression identity.

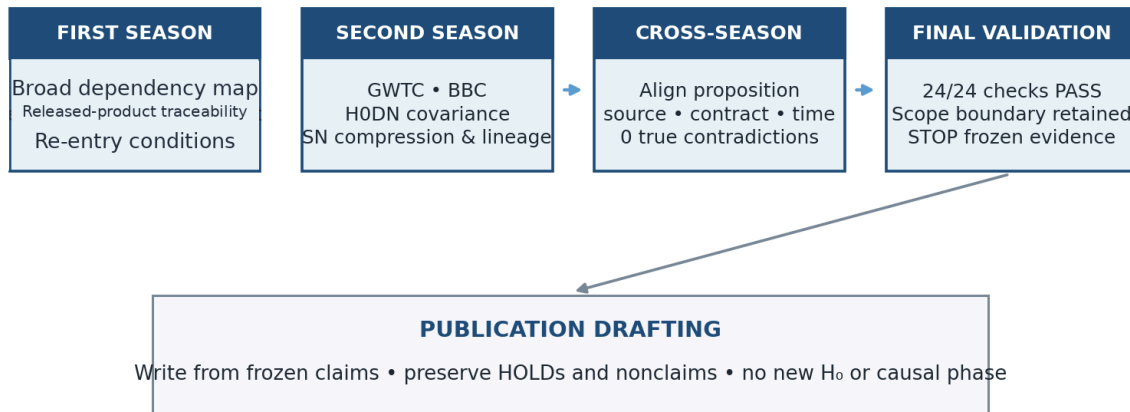


Figure 1. Audit progression and stopping boundary. The first season mapped dependencies; the second season tested selected public products; the cross-season audit aligned propositions and scopes; the final package retested the highest-information mechanisms. Publication drafting follows the STOP decision and does not reopen scientific analysis.

3. A seven-coordinate working model of reproducibility

The final validation record separates seven questions that are often conflated:

F0 — Artifact identity. Are the same bytes, version, commit, and release record being examined?

F1 — Numerical/output traceability. Can a stated table entry, quantile, or headline output be recovered under a fixed numerical contract?

F2 — Algebraic representation relation. Are two representations exactly related, invertible, or deliberately lossy?

F3 — Target-inference invariance or sufficiency. Does the transformation or compression preserve the likelihood, estimate, and uncertainty for the declared target?

F4 — Diagnostic/model-support sufficiency. Does the retained object preserve residual, goodness-of-fit, covariance-support, and model-checking information?

F5 — Executed lineage/provenance. Can the actual run connect its input bytes, configuration, intermediate products, and final reported row or posterior?

F6 — Generative/causal closure. Does the evidence support the data-generating model, correction validity, physical cause, or causal attribution?

F2 and F3 must be separate. In the HODN case, an invertible row transformation gives an exact algebraic correspondence (F2). An on-support residual retains the same target-dependent singular-Gaussian quadratic under such a transformation, but the rounded public vector is off support and the adopted projected Moore–Penrose loss does not preserve the target under the tested non-orthogonal transformation (F3). In the supernova case, compression from 277 rows to one scalar is lossy (F2), yet it preserves the frozen fixed-covariance, one-intercept target (F3) while failing to preserve residual diagnostics (F4). These examples are not edge cases in terminology; they change what a reader may legitimately infer from a successful numerical reproduction.

Evidence coordinates are question-specific; they are not a cumulative score

	F0 Artifact	F1 Output	F2 Relation	F3 Target	F4 Diagnostic	F5 Lineage	F6 Causal
TDCOSMO	PASS	PASS	PASS	PASS	N/T	HOLD	HOLD
GWTC v1	MIXED	PASS	PASS	PASS	N/T	HOLD	HOLD
Local ladder / BBC downstream	PASS	PASS	PASS	N/T	MIXED	HOLD	HOLD
H0DN covariance	PASS	PASS	PASS	FAIL	HOLD	MIXED	HOLD
SN compression	PASS	PASS	MIXED	PASS	FAIL	HOLD	HOLD
SN same-name contrast	PASS	PASS	N/T	N/T	MIXED	HOLD	HOLD
DESI atlas	PASS	PASS	PASS	MIXED	N/T	HOLD	HOLD

Not tested
Failed stated test
Hold/open
Mixed/partial
Pass

Figure 2. Seven-coordinate evidence matrix for the principal audited cases. Colors encode PASS/strong support, MIXED or partial support, HOLD/open, FAIL for the stated test, and not tested. A row is not a score: later coordinates cannot be inferred from earlier ones. Statuses follow the accompanying cross-season evidence-coordinate matrix.

4. First-season dependency map

The first season remains the broadest map of the project. Its role after the second season is not “independent confirmation of every branch,” because the second season directly revisited only GWTC, Pantheon+ BBC, H0DN singular covariance, and H0DN supernova diagnostics. Instead, the first season supplies dependency structure, frozen numerical anchors, and explicit re-entry requirements. Its observational source coverage includes primary DESI, Planck, ACT, SPT-3G, Megamaser Cosmology Project, and peculiar-flow records [5–10].

Table 1. First-season branch clusters and current bounded readings.

Cluster	Audited object	Established within scope	Boundary retained after cross-season audit	Re-entry evidence
Local ladder / BBC	Released corrected vector and structural decompositions	A recorded fixed-effect slope was algebraically localized to the explicit BBC term plus a near-zero bias-removed proxy	No truth-level closure, BBC overcorrection, or causal defect	Matched truth, fit, pre/post vectors, covariance, and run manifest
BAO–BBN bridge	Frozen H_0 (r,d) contract	Approximate -6.8% local- H_0 sound-horizon budget	Contract-dependent budget, not new physics	Matched early-model likelihood or profile products
CMB–DESI geometry	Released constraints and response directions	Near-common Ω_m – h (r,d) direction described	Shared-source geometry is not an independent causal explanation	Joint simulations, cross-CMB covariance, full posteriors
DESI influence atlas	Same released vector under mode decompositions	MIDZ(COMMON) _(ISO) was a high-leverage released-vector mode; LRG1(DM) _(I) provided a counter-anchor	Multiple bases are not multiple observations or raw-pipeline tests	Aligned raw variants and joint/cross-fit covariance
ACT variants	Official public chains	Tested centers were stable within a correlated-source variant set	Not a complete component-likelihood audit	New independent product or exact component likelihood

Cluster	Audited object	Established within scope	Boundary retained after cross-season audit	Re-entry evidence
TDCOSMO	Public chain products	Alternate implementation recovered 13/13 structures, 39/39 quantiles within tolerance, and 12/12 Table 6 rows at published precision	Output-level traceability only; not likelihood, sampler, or astrophysical-model reproduction	Original likelihood, sampler configuration, diagnostics, and portable environment
Local flow / MCP-CF4	Public route interface	Single-galaxy dominance was rejected within the frozen tests; dependencies were mapped	Joint-flow inference remains open	Distance samples, joint covariance, zero-point cross-covariance, likelihood
HODN network	Public data, design, covariance, and code contract	Baseline and frozen-representation solver behavior were recovered	Correlated peculiar-velocity and covariance generation remain incomplete	Official source matrix/removal vector or author package
Cross-route synthesis	Non-ladder routes	Shared assumptions and data were mapped	No naive averaging of dependent branches	Compatible likelihood contracts and cross-route covariance

The first-season map therefore supports triage. It identifies where more arithmetic on the same released vector has low information gain and where a new external product would materially change the evidence state. It does not support combining branch summaries as though they were independent measurements.

5. Focused public-product audits

5.1 GWTC-4 and GWTC-5 v1: headline recovery is not metric provenance

The standard-siren branch fixed two official posterior records and their files. For GWTC-4, the audited posterior had unrounded percentiles

$$q_{(15.865)}=67.1096877144, q_{50}=76.6372575526, q_{(84.135)}=89.6794449380.$$

The specified display rule gives $76.6^{(+13.0)(-9.5)} \text{ km s}^{-1} \text{ Mpc}^{-1}$, matching the frozen publication statement [13,14]. For GWTC-5 v1, the corresponding percentiles were 63.8759189462, 71.0170052237, and 80.0024959217, yielding $71.0^{(+9.0)(-7.1)} \text{ km s}^{-1} \text{ Mpc}^{-1}$, again matching the frozen v1 statement [15,17]. A custom implementation and a standard type-7 implementation agreed for all six percentiles.

Using the average of the upper and lower 68.27% half-widths for those two audited headline posteriors gives widths 11.2848786118 and 8.0632884877 and a relative reduction of 28.5478491605%. This is a diagnostic of the audited pair. It is not a reproduction of the 25.7% reduction printed in GWTC-5 v1, because the exact old-side comparator used by the publication's metric was not closed by the public provenance inspected in the branch. The correct conclusion is two-part: the v1 headline quantiles were traceable, while the historical reduction metric remained on HOLD.

Artifact identity matters here in a literal temporal sense. On 4 August 2026, after the frozen v1 audit, the GWTC-5 arXiv record was revised to v2. The GWTC-5 v2 abstract reports $71.7^{(+9.4)(-7.5)} \text{ km s}^{-1} \text{ Mpc}^{-1}$ and a 22.0% uncertainty reduction [16]. The v2 values do not invalidate the audit; they are a different versioned proposition. This manuscript therefore refers to 71.0 and 25.7% only as GWTC-5 v1 quantities tied to the frozen Zenodo product, not as current headline results.

5.2 Pantheon+ BBC: a completed source-readiness HOLD

BBC is a simulation-based framework for correcting selection effects and contamination in Type Ia supernova Hubble diagrams [11]. The audit fixed the Pantheon+ data-release repository at commit c447f0fea703fcd0fff57de5000947b5ca81286b [12]. At that commit, public files establish the final real-data vector, corrected-magnitude fields, covariance products, ten data-sized simulation realizations, and PIPPIN configuration families. They do not uniquely assemble the bundle needed for a truth-level closure:

BiasCor truth → matched fitted sample → BBC pre/post vector → same-order covariance → executed configuration.

Several PIPPIN paths resolve to project-local or environment-dependent locations, multiple BiasCor families are present, and the approximately 2 GB BiasCor training files referenced by the public README were not supplied as a versioned, checksummed public bundle in the frozen state. The branch did not substitute the ten public test realizations for the nominal training products and did not reuse final real-data covariance as simulated closure covariance.

This is a positive audit result about source readiness: the available public state was insufficient to define a unique truth-level closure test. It is not evidence that BBC corrections are wrong, too large, or responsible for the Hubble tension. The first-season downstream algebraic localization remains valid for its released-vector question; the second-season HOLD concerns a different, upstream generative question.

5.3 HODN singular covariance: stable arithmetic and an off-support projected-loss test

The HODN public baseline is $H_0=73.50\pm0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1]. Under the frozen public code and data contract, the audit reproduced the more precise numerical values

$$H_0=73.49875364360662\pm0.8088000253378453 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

with $\chi^2=117.55968168211207$ and adjusted degrees of freedom 119. The 255 by 255 covariance was positive semidefinite under the material-eigenvalue rule, with rank 183 and nullity 72; the 255 by 64 design matrix produced a full-rank 64 by 64 normal system. Within the original row representation, changing the absolute pseudoinverse cutoff from 10^{-14} through 10^{-7} produced zero recorded spread in H_0 .

The support distinction is essential. If a symmetric positive-semidefinite covariance factors as $C=BB^T$ and a residual lies on support, $r=Bu$, then $r^TCr=u^Tu$. After any invertible row transformation D , the transformed residual is DBu and the transformed quadratic has the same value. Thus the target-dependent quadratic of the on-support singular Gaussian is invariant under invertible coordinate changes.

The rounded public HODN vector is not in that setting. The covariance null-space projector P_0 satisfies $\|P_0A\| \approx 1.04\times10^{-13}$, while $\|P_0y\|_2=0.1887490826897 \text{ mag}$. The design is nearly in covariance support, but the rounded data vector has an off-support component that cannot materially be removed by varying the parameters. The audited Moore–Penrose objective therefore acts as a projected loss: it ignores the Euclidean null-space component. That off-support extension is not generally invariant because a non-orthogonal transformation changes the Euclidean projection geometry.

The representation test applied an invertible positive diagonal row scaling D to all three objects: $(A,y,C) \mapsto (DA,Dy,DCD^T)$. Under the projected Moore–Penrose loss used for this off-support vector, the audited target shifted by

$$\Delta H_0=-0.052445422611000936 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Four project-internal linear-algebra paths—SciPy pseudoinverse, explicit SVD, symmetric eigendecomposition, and retained-support whitening followed by rank-revealing least squares—agreed on the shift to a spread of $1.99 \times 10^{-12} \text{ km s}^{-1} \text{ Mpc}^{-1}$. An 80-digit solve gave $-0.05244542261171722333\dots$, and dense orthogonal transformations preserved H_0 to a maximum absolute change of $1.51 \times 10^{-10} \text{ km s}^{-1} \text{ Mpc}^{-1}$. An exact two-dimensional rational fixture reproduces the same off-support projected-loss mechanism (Technical Supplement A). The 72-dimensional null space matches the dimension $(37-1)(3-1)=72$ of a host-by-anchor interaction subspace in the audited decomposition. This dimensional and numerical localization does not identify why the public covariance has this structure.

None of these findings supplies a corrected H_0 . The exploratory full-rank variance-component example in the branch was explicitly non-canonical; its conditional H_0 was 73.4943 rather than a proposed replacement. The established conclusion is narrower: the tested off-support projected Moore–Penrose loss is not invariant under the exact non-orthogonal transformation. It does not show that arbitrary singular Gaussian inference is non-invariant, and a literal degenerate-Gaussian reading of the rounded public vector requires further generative clarification.

5.4 Supernova scalar compression: target sufficiency versus diagnostic sufficiency

The HODN network uses a supernova Hubble-flow component connected to public Pantheon+ products [2–4]. The audit identified a 277-row block and analyzed only the fixed-covariance, one-intercept linear Gaussian model. Let d be the 277-vector, C its fixed covariance, and 1 the intercept design. Then

$$\hat{a} = (1^T C^{-1} d) / (1^T C^{-1} 1), V = (1^T C^{-1} 1)^{-1}.$$

For the frozen model, $\hat{a} = 0.7163834210954622$ and $\sqrt{V} = 0.0018926416391806472$. Completing the square gives

$$(d - a1)^T C^{-1} (d - a1) = \chi^2_{\min} + ((a - \hat{a})^2) / (V).$$

Consequently, $(\hat{a}, V)$ is sufficient for the likelihood dependence on the single target a under that fixed model and covariance; substituting the scalar block into the frozen network reproduces the same target estimate and uncertainty. But the scalar pair omits $\chi^2_{\min} = 206.7606364373241$ for 276 degrees of freedom. It is therefore not sufficient for assessing residual magnitude or pattern. The final validation checked the identity over seven profile points with maximum numerical residual 3.41×10^{-13} and constructed exact synthetic vectors with the same compressed mean and variance but residual chi-squares 2 and 8. Parameter sufficiency and diagnostic sufficiency are logically distinct [22–25,27].

5.5 Residual localization and the same-CID evidence frontier

The omitted supernova residual term was decomposed into orthogonal subspaces. Thirty names had multiple rows, comprising 69 of the 277 rows and 39 within-name contrast degrees of freedom. The total residual split as

$$206.760636437324 = 11.209315063603 \text{ (39 df)} + 195.551321373699 \text{ (237 df)}.$$

Under the registered beta-tail calibration, the within-name statistic had two-sided probability $1.8736724464562464 \times 10^{-4}$. The low contrast statistic persisted under the frozen covariance alternatives: the full case gave 11.209315/39, removal of rowwise velocity terms gave 14.734236/39, and STATONLY gave 16.233448/39 with $p = 4.856832550848106 \times 10^{-4}$. The canonical Phase 1C package included an alternate null basis, 32 orthogonal trials, and a separate project-internal parser/eigendecomposition check. The final validation reran all 24 Phase 1C gates and the separate verifier successfully.

This result localizes where an unusual residual feature lies; it does not identify its cause. Public lineage work mapped all 277 official Hubble-flow rows and all 76,729 covariance elements exactly. For the 69 multirow

entries, Phase 1D identified unique public photometric input candidates for 38 rows; a target-excluded survey crosswalk resolved candidates for the remaining 31, giving 69/69 candidate coverage within the frozen seven-directory universe. At the next layer, 0/48 cross-series pairs had byte-exact reused observation lines, 4/48 met the registered numeric-payload compatibility rule, 434/434 used-filter records mapped to public transmission assets, all 6,744 observations were covered, and 7/7 series had configuration-level anchors.

These are substantial F0–F5 partial results, but the public evidence did not close the exact executed route

Raw photometry → light-curve fit → BiasCor/BBC processing → final corrected-magnitude network row.

Accordingly, the audit does not infer duplicated observations, statistical independence or dependence of specific rows, a survey cause, a row-removal prescription, an overestimated covariance, or a corrected network result.

5.6 TDCOSMO: bounded output-level implementation diversity

The TDCOSMO branch provides a contrasting successful output trace. A project-internal alternate implementation was first tested blindly on three sentinel files and then frozen before extension to twelve flat- Λ CDM Table 6 chains plus one release-identifier product. It passed 13/13 structural comparisons, 39/39 q16/q50/q84 comparisons within preregistered tolerance, and 12/12 Table 6 rows at published precision, using public commit `d7f38db341f68be1df0d9ac1fc528c45113f94cf` and the registered equal-weight type-7 rule [18].

The result is strong evidence for artifact and output traceability and for implementation diversity at the endpoint. It is not reproduction of the original likelihood evaluation, sampler behavior, convergence assessment, blinding process, or astrophysical generative model. This bounded success demonstrates why a high F1 state need not imply high F5 or F6.

6. Cross-case synthesis

Table 2. Principal distinctions established by the audited cases.

Case	What passed	What did not follow	Highest-value unresolved object
TDCOSMO chains	Structure, quantiles, printed Table 6 values	Original likelihood/sampler or astrophysical validation	Portable likelihood and sampling environment
GWTC-4/5 v1	Six headline percentiles and displayed intervals	Provenance of the v1 25.7% historical metric	Exact old comparator and mapping
Pantheon+ BBC	Final-vector identity and downstream algebraic localization	Truth-level correction closure or overcorrection claim	Versioned truth→fit→pre/post→covariance bundle
HODN covariance	Baseline, rank, cutoff stability, and off-support projected-loss test	General non-invariance, corrected H_0 , or physical cause	Covariance generative model, unrounded inputs, and external replication
SN compression	One-intercept target likelihood	Residual adequacy or pattern	Retained residual diagnostics for the question asked
Same-CID lineage	Residual localization and extensive public candidate mapping	Executed ancestry, dependence, or cause	Exact run manifest and final-row ancestry

Three general observations follow within the audited record.

First, output traceability is version-specific. The GWTC-5 revision demonstrates that a valid audit can become historically scoped without becoming false. A reproducible claim must carry the file, release, quantile rule, and time.

Second, sufficient statistics are target-specific. The supernova scalar is sufficient for one parameter under a fixed Gaussian model but insufficient for residual diagnostics. A phrase such as “the compression is sufficient” is incomplete unless the target and model-checking task are named.

Third, support status controls the invariance statement. A singular-Gaussian quadratic evaluated strictly on covariance support is invariant under invertible row coordinates. The audited HODN vector instead has an off-support component and is evaluated through a projected Moore–Penrose loss. Because that projection is defined by ambient Euclidean orthogonality, a non-orthogonal congruence can change the projected extension. The observed shift is not ordinary solver instability: several algorithms and high-precision arithmetic agree. It is specific to the tested off-support projected-loss prescription.

Across all cases, a source-readiness HOLD can be a completed result. When public products do not define a unique test, stopping is more informative than silently choosing an undocumented substitute. The hold records specify which new object would change the state and prevent repeated low-information reanalysis.

7. Final internal validation and closure

The final validation program admitted only checks capable of changing a manuscript-level claim. It did not reopen broad branch searches. For HODN, it tested four solver paths, high-precision normal solves, orthogonal transformations, a non-orthogonal congruence defect, covariance-support projections, and an exact rational fixture. For the supernova block, it used a separate parser, Cholesky and eigendecomposition routes, complete-square identities, and exact synthetic equal-compression/unequal-residual examples. For same-CID contrasts, it reran the canonical verifier rather than inventing a new statistic after the result was known.

All 24 registered numerical checks passed. The four HODN off-support projected-loss transformation paths had a spread of 1.9895×10^{-12} , the maximum orthogonal-transform shift was 1.5085×10^{-10} , and the relative Frobenius defect of the naive Moore–Penrose congruence relation was 3.80468×10^{-4} . The projected design and data norms were 1.0381×10^{-13} and 0.1887491, respectively. The same-CID Phase 1C gates passed 24/24 in their canonical package.

The resulting STOP applies to further internal reanalysis of the current frozen evidence. It does not prohibit future work if a named re-entry condition is met—for example, an external HODN replication, an official BBC truth-closure bundle, exact GWTC comparator provenance, an executed supernova lineage manifest, DESI raw variants with cross-fit covariance, local-flow likelihood products, or a portable TDCOSMO generation environment. Publication drafting is downstream of the STOP and does not constitute a new scientific phase.

8. Discussion

8.1 Reproducibility should be reported as a vector, not a badge

Reproducible computational research requires inspectable objects and an executable link between claims and evidence [28]. The cases here suggest that a single badge compresses away essential distinctions. At minimum, a reader should know whether the same artifact was used, whether the output was recovered, whether the target inference survived transformation or compression, whether diagnostics were retained, and whether the executed and generative paths were closed.

The seven-coordinate model is deliberately practical. It can be recorded in a table and linked to source locators. It also discourages upward inference: an F1 success cannot be promoted to F6 without new evidence. Conversely, a later-coordinate HOLD does not erase an earlier-coordinate success. TDCOSMO output traceability remains

valuable even though the sampler was not reproduced; HODN baseline recovery remains correct in the frozen representation even though the off-support projected-loss extension is representation-sensitive under the tested scaling.

8.2 Residual information is part of the scientific object

Parameter compression can be exact for a declared target, but model assessment asks a different question. Classical work on sufficient statistics and residual examination already separates estimation from diagnostic information [22–25]. The supernova case gives a concrete cosmology example: the compressed scalar carries the curvature and location of the fixed-covariance, one-intercept likelihood, while the omitted constant contains the entire residual goodness-of-fit for that block. If a downstream user receives only the scalar, the target estimate can be right even though an unexpected residual structure is invisible.

This distinction does not make compression improper. It specifies what should accompany compression when the public purpose includes validation: at least the minimized residual term, degrees of freedom, and enough projected residual information to test salient patterns. The same-name subspace result further shows that an aggregate residual can look unremarkable while a scientifically relevant subspace behaves unusually.

8.3 Singular covariance requires a declared geometry

In full-rank generalized least squares, invertible row transformations preserve inference when covariance transforms by congruence. The same target-dependent quadratic invariance holds for residuals strictly on the support of a singular Gaussian [19–21]. The qualification arises when an observation is off support and the Moore–Penrose quadratic is extended to it as a projected loss: the ambient Euclidean projection is then part of the prescription. The HODN example shows that solver agreement and cutoff stability do not settle this off-support representational question.

The magnitude of the observed shift is small relative to the published HODN uncertainty and is not presented as a tension-relevant correction. Its importance is logical and limited: two exact coordinate descriptions of the tested off-support rounded vector need not yield identical targets under the adopted projected-loss extension. A publication using such a convention should state the representation, support interpretation, invariance class, and treatment of off-support observations. The result should not be generalized to arbitrary singular Gaussian inference.

8.4 Provenance claims should stop at the executed frontier

The BBC and same-CID audits show why public repository richness is not equivalent to executed lineage. A repository can contain final vectors, test simulations, filter curves, and configuration templates while still lacking a unique mapping to the production run. Causal language—“this correction caused the shift,” “these rows reuse the same observations,” or “this pipeline is defective”—requires evidence that bridges the executed chain. Filename similarity, compatible numeric payloads, or plausible configuration families are useful candidate evidence but do not complete that bridge.

8.5 Negative and bounded results reduce future search cost

The project’s HOLD and STOP states are not failures to continue. They record where the expected information gain from further manipulation of unchanged public products became low, and they define the external artifact that would make re-entry rational. This converts an open-ended audit into a falsifiable queue of evidence requirements.

9. Limitations and explicit nonclaims

This is a project-level internal audit of public products, not an external replication by an unaffiliated team. The same author directed the audit and manuscript synthesis, and AI systems assisted under human review. Internal implementation diversity reduces some coding risks but does not supply social or institutional independence.

The first-season map is broad but heterogeneous. Not every branch was revisited in the second season, and released-vector decompositions are not substitutes for raw-pipeline tests. The evidence coordinates are qualitative working labels induced by these cases rather than a validated measurement scale or universal theory. The HODN transformation result is specific to the frozen matrices, rounded off-support vector, tested scaling, and projected Moore–Penrose loss. It does not establish non-invariance for an on-support singular Gaussian likelihood. The supernova sufficiency result is specific to the fixed-covariance, one-intercept linear Gaussian model. Tail probabilities in the same-CID branch diagnose the registered statistic under its calibration; they do not by themselves identify a data-generating cause.

The manuscript does not establish any of the following:

- a corrected, preferred, or replacement value of H_0 ;
- a revised statistical significance for the Hubble tension;
- a unique astrophysical, instrumental, survey, or software cause;
- that Pantheon+ BBC corrections are excessive or incorrect;
- that the HODN public value or covariance is “wrong”;
- a defective named pipeline, a row-deletion rule, or a covariance correction;
- a solution to the Hubble tension or evidence for new physics;
- external replication of the project’s reimplementations.

10. Conclusions

The completed audit supports a layered reading of public Hubble-constant inference products. Headline numbers can be reproducible while historical metric provenance remains open; an off-support rounded vector evaluated by a projected Moore–Penrose loss can yield representation-dependent estimates even though the on-support singular-Gaussian quadratic is invariant; and a scalar can be sufficient for a fixed-model target parameter while discarding all information needed for residual assessment. Public candidate inputs can be mapped extensively without establishing which executed run generated a final corrected row.

No genuine scientific contradiction was found between the two audit seasons after proposition, source, contract, and time were aligned. The strongest supported result is therefore not a new cosmological value but a clearer evidence boundary. Artifact identity, output traceability, representation, target sufficiency, diagnostic sufficiency, executed lineage, and causal closure should be reported separately. The current frozen internal program is closed. Further scientific re-entry requires a specific new external product or replication, not another analysis of the same bytes.

Data and code availability

The accompanying Publication Reproducibility / Evidence Supplement, version 1.1.0 (10 August 2026), identifies the internal source packages, result keys, and frozen external records used for principal numbers, tables, and figures. It contains project-created code, compact results, provenance records, and integrity manifests; third-party raw data and posterior products are not redistributed. Retrieval identifiers, fixed commits, filenames,

and hashes are supplied instead. No new public repository, DOI, or endorsement is asserted. The four project master packages remain the authoritative internal record.

AI assistance disclosure

Artificial intelligence systems assisted with audit design support, source inspection, literature triage, document organization, code generation, project-internal reimplementation, numerical cross-checking, archive assembly, evidence synthesis, consistency review, and manuscript drafting. AI systems are not authors, subject-matter experts, external replicators, or independent validators. Keiji Yoshimura reviewed the outputs and retains responsibility for source selection, scientific judgments, scope decisions, interpretation, and the manuscript. Project-internal AI-assisted reimplementations are not represented as external replication.

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